
2026 Chantilly Math Competition

Elementary School Division - Team Round Solutions

1. (9 pts) Given that

$$f(x) = 7x + 5,$$

compute $f(5)$.

Madhavan Kandagatla

Solution

$$f(5) = 7(5) + 5 = \boxed{40}.$$

2. (9 pts) 2 trains are on the same track traveling towards each other. 1 train travels at a speed of 11 mph and the other travels at a speed of 12 mph. When the 2 trains are 2024 miles apart, a bird begins to fly back and forth between the 2 trains at a constant speed of 13 mph. How many miles will the bird travel before the trains meet?

Rishi Jagarlapudi

Solution

$$13 \cdot \frac{2024}{11 + 12} = \boxed{1144}.$$

3. (10 pts) An equilateral triangle and regular hexagon have the same area. The regular hexagon has side length 1. Let p_t and p_h be the perimeter of the equilateral triangle and the perimeter of the regular hexagon, respectively. Compute $p_t^2 + p_h^2$.

Aryan Raj

Solution The side length of the equilateral triangle will be $\sqrt{6}$ times the side length of the regular hexagon because a regular hexagon can be tiled with 6 equilateral triangles. Therefore, our answer is $(3\sqrt{6})^2 + 6^2 = 54 + 36 = \boxed{90}$.

4. (10 pts) Given $xy + yz = 9$, $yz + zx = 21$, and $zx + xy = 24$, compute $|xyz|$.

Aryan Raj

Solution $xy + yz + zx = \frac{9+21+24}{2} = 27$

$$xy = 27 - 21 = 6$$

$$yz = 27 - 24 = 3$$

$$zx = 27 - 9 = 18$$

$$|xyz| = \sqrt{6 \cdot 3 \cdot 18} = \boxed{18}$$

5. (11 pts) A number is a super-prime if it is both prime and has the sum of its digits also prime. What is the smallest super-prime that has the sum of its digits not be a super-prime?

Aidan Shin

Solution Note that 2, 3, 5, 7, and 11 are all super-primes, so any such number that meets the conditions we want must have its sum be at least 13. We can also see that the units digit is always odd, meaning the tens digit must be even. This narrows down our cases a lot, and we can simply test primes until we get that $\boxed{67}$ is prime and its sum of digits is 13, which is not a super-prime, so we are done.

6. (11 pts) Moe has an empty multiplication table with 9 rows and 9 columns. Moe fills every cell of the table by writing xy in the cell in the x th row and the y th column. What is the sum of all 81 numbers Moe writes?

Aryan Raj

Solution

$$(1 + 2 + \dots + 9)^2 = 45^2 = \boxed{2025}.$$

7. (12 pts) What 2-digit positive integer do you get when you double the answer to this problem, reverse its digits, and add 1?

Aryan Raj

Solution Let the 2-digit answer to the problem be $10a + b$.

Then, doubling gives $2(10a + b) = 10(2a) + (2b)$.

If $b < 5$, reversing the digits gives $10(2b) + (2a)$.

Adding 1 must give the answer to the problem.

Therefore, $10(2b) + (2a) + 1 = 10a + b$, so $19b + 1 = 8a$.

It is easy to see that this has no solutions.

Therefore, $b \geq 5$, so reversing the digits gives $10(2b - 10) + (2a + 1)$.

Adding 1 must give the answer to the problem.

Therefore, $10(2b - 10) + (2a + 1) + 1 = 10a + b$, so $19b - 98 = 8a$.

It is easy to see that the only solution is $a = 2$ and $b = 6$.

Therefore, the answer is $10(2) + 6 = \boxed{26}$.

8. (12 pts) A rhombus with perimeter 100 rests flat on one of its sides. The height of the rhombus is 24. Compute the sum of the lengths of the rhombus' 2 diagonals.

Aryan Raj

Solution

$$\sqrt{24^2 + \left(\frac{100}{4} - \sqrt{\left(\frac{100}{4}\right)^2 - 24^2}\right)^2} + \sqrt{24^2 + \left(\frac{100}{4} + \sqrt{\left(\frac{100}{4}\right)^2 - 24^2}\right)^2} = \boxed{70}.$$

9. (13 pts) John wants to buy n cheeseballs. He can only buy cheeseballs in packs of 7 or 4. For how many positive integer values of n can John NOT buy EXACTLY n cheeseballs?

Riyan Nayak

Solution By the Chicken McNugget Theorem, $7 \cdot 4 - 7 - 4 = 17$ is the highest number of cheeseballs that CAN'T be bought. This can also be seen by observation. Now we have to check the positive integers from 1 to 17. All numbers divisible by 4 and 7 obviously work, ruling out $4 + 2 = 6$ options. $4 + 7 = 11$ and $4 + 4 + 7 = 15$ work ruling out 2 more options. This leads to a total of $17 - 6 - 2 = \boxed{9}$ numbers of cheeseballs that John is unable to purchase.

10. (13 pts) A quadrilateral has its vertices at coordinates $(-5, 0)$, $(-3, 4)$, $(3, -4)$, and (a, b) . The 4 sides have midpoints of $(-4, 2)$, $(-1, -2)$, $(8, 7)$, $(11, 3)$ in some order. Compute $6a + 7b$.

Aryan Raj

Solution

$$6 \cdot 19 + 7 \cdot 10 = 114 + 70 = \boxed{184}.$$

11. (14 pts) Arsenii, Justin, Rishi, and 6 other Math Club members line up for a group photo. Arsenii, Justin, and Rishi act silly whenever any 2 of them are adjacent in the lineup. How many ways are there to line up all 9 Math Club members such that Arsenii, Justin, and Rishi all don't act silly?

Aryan Raj

Solution Call Arsenii, Justin, and Rishi fellas for brevity. Let there be a people before the first fella, b people between the first and second fella, c people between the second and third fella, and d people after the third fella. Then, we must have $a + b + c + d = 6$ where $b, c \geq 1$. This has $\binom{7}{3}$ solutions. Each solution gives $3!6!$ lineups, because we need to permute fellas and the other members. The answer is therefore $\binom{7}{3}3!6! = \boxed{151200}$.

12. (14 pts) Let $S(n)$ denote the sum of the digits of a positive integer n .

What is the smallest positive integer x such that 13 divides both $S(x)$ and $S(x + 1)$?

Aryan Raj

Solution We claim that

$$S(x + 1) - S(x) = 1 - 9(\text{number of 9s } x \text{ ends with}).$$

If x doesn't end in 9, then clearly $S(x + 1) - S(x) = 1$, satisfying our claim.

Otherwise, $x = 10y + 9$ and $x + 1 = 10(y + 1)$, so $S(x) = S(y) + 9$ and $S(x + 1) = S(y + 1)$.

Therefore, $S(x + 1) - S(x) = S(y + 1) - S(y) - 9$. By induction, our claim is proven.

We need 13 to divide $1 - 9(\text{number of 9s } x \text{ ends with})$.

Therefore, the smallest possible number of 9s is 3.

Since $3 \cdot 9 = 27$, we need to add 12 more.

Therefore, the answer is $\boxed{48999}$.

13. (15 pts) There is an empty 3×3 grid. In every square, a distinct positive integer divisor of 36 is written. The product of the 3 numbers along any row, column, or diagonal of the grid is always the same. Compute the sum of the numbers in the 4 corners of the grid.

Aryan Raj

Solution The product of all positive integer divisors of 36 is 6^9 .

Therefore, the product of the 3 numbers along any row, column, or diagonal is $6^3 = 216$.

Let the middle number in the grid be m .

Then, $(\frac{216}{m})^4 = \text{product of 8 non-middle numbers} = \frac{6^9}{m}$, so $m^3 = 6^3$ so $m = 6$.

Then, the numbers in the grid must be

$\frac{6}{x}$	$6xy$	$\frac{6}{y}$
$\frac{6x}{y}$	6	$\frac{6y}{x}$
$6y$	$\frac{6}{xy}$	$6x$

for some distinct x and y .

WLOG, we can assume x and y both divide 6.

Furthermore, neither x nor y can be 1, or else 6 will appear twice.

In addition, xy must divide 6, so that $\frac{6}{xy}$ is a positive integer.

Therefore, x and y are 2 and 3 in some order.

Therefore, the sum of the numbers in the 4 corners of the grid is

$$\frac{6}{x} + \frac{6}{y} + 6x + 6y = \frac{6}{2} + \frac{6}{3} + 6(2) + 6(3) = 3 + 2 + 12 + 18 = \boxed{35}.$$

14. (15 pts) An ant is on a vertex of a cube. Every second, it moves to a randomly chosen adjacent vertex. What integer is nearest to 100 times the probability that the ant visits the vertex of the cube farthest from its starting point before returning to its starting point for the first time?

Aryan Raj

Solution Let $p(n)$ be probability that, starting from a vertex n away from the starting vertex, the ant visits the vertex of the cube farthest from its starting point before returning to its starting point for the first time.

$$p(0) = p(1)$$

$$p(1) = \frac{2}{3}p(2)$$

$$p(2) = \frac{2}{3}p(1) + \frac{1}{3}p(3)$$

$$p(3) = 1$$

$$\text{Substitute to get } p(2) = \frac{4}{9}p(2) + \frac{1}{3} \text{ so } p(2) = \frac{3}{5}.$$

$$\text{Therefore, } 100p(0) = 100p(1) = 100 \cdot \frac{2}{3} \cdot p(2) = 100 \cdot \frac{2}{3} \cdot \frac{3}{5} = \boxed{40}.$$

15. (16 pts) There exists a prime p and positive integers a , b , and c , such that

$$\frac{pa+1}{2} + \frac{pb+2}{3} - \frac{pc+3}{5}$$

is an integer divisible by p . Compute the smallest possible value of $p+a+b+c$.

Aryan Raj

Solution We have

$$\frac{pa+1}{2} + \frac{pb+2}{3} - \frac{pc+3}{5} = pk$$

for some integer k .

Multiplying by 30 gives $15pa+15+10pb+20-6pc-18 = 30pk$, so $p(15a+10b-6c-30k) = -17$.

Since 17 is prime, $p = 17$, so $15a + 10b - 6c - 30k = -1$.

Plugging in $a = b = c = 1$ gives $15a + 10b - 6c = 19$.

Plugging in $a = c = 1$ and $b = 2$ gives $15a + 10b - 6c = 29 = 30(1) - 1$.

Therefore, the answer is $p+a+b+c = 17+1+2+1 = \boxed{21}$.

16. (16 pts) Carl has a triangular piece of cheese. Whenever Carl cuts a piece of cheese, he cuts it along a straight line, splitting it into 2 pieces of cheese. Carl always cuts one piece at a time. After several cuts, Carl has 21 pieces of cheese, and the total number of sides of all these pieces of cheese is 67. Every cut passed through 0, 1, or 2 existing vertices of the piece of cheese it cut. What is the total number of vertices passed through?

Aryan Raj

Solution Let a , b , and c cuts pass thru 0, 1, and 2 vertices, respectively.

For generality, let Carl end with p pieces of cheese and t total sides.

Every cut increases number of pieces by 1, so $a+b+c = p-1$.

Cutting thru a vertex turns 1 vertex into 2 vertices.

Cutting thru an edge turns 0 vertices into 2 vertices.

Therefore, since we end with t total vertices, $4a+3b+2c = t-3$.

Subtracting 2nd equation from 4 times 1st equation gives $0a+1b+2c = 4p-t-1$.

Plugging in $p = 21$ and $t = 67$ gives $4(21) - 67 - 1 = \boxed{16}$.

- TB. (Tiebreak) Ryan has a random number generator with 2 integer parameters $1 = m < M = 67$. While $M - m > 1$, the random number generator picks some integer between m and M exclusive, and Ryan changes either m or M , each option with probability $\frac{1}{2}$, to that integer. Estimate the integer nearest to 1 billion times the probability that $m = 41$ when $M - m = 1$.

Aryan Raj

Solution Let $A(x, y)$ be the answer when starting with $m = x$ and $M = y$.

When $y - x = 1$, $A(x, y) = 1$ if $x = 41$ and $A(x, y) = 0$ otherwise.

Otherwise,

$$A(x, y) = \frac{1}{y - x - 1} \sum_{k=x+1}^{y-1} \frac{A(x, k) + A(k, y)}{2}.$$

Coding up this recursion gives

$$A(1, 67) = \frac{424689107887844190802240558129596195}{42535295865117307932921825928971026432} \approx .009984392943558356.$$

Therefore, the integer nearest to $10^9 A(1, 67)$ is 9984393.